

SE 461 Software Testing and Quality (Spring 2026)

Group Project: Test Design and Development

Due on Sunday, May 3, 11:59PM.

In this project we will work in groups to test a real software system using some the testing methods and tools covered in SE 461. The subject system is a video game called Snake (written in Java), and its source code is included in the SnakeRemake-Testing.zip file attached to this file. Your tasks of this project will include:

- 1) **Project preparation:** install all the tools (e.g., Eclipse, JaCoCo mentioned below) that you may need for this project in your computer. Download the provided Snake source code and import it into your Eclipse environment. Run the code, go through the code, and understand its overall logic and structure.
- 2) **Test design:** Use one or more of the test coverage criteria that we have learned to design your test cases. First, choose a test criterion that you think is appropriate for this project. Then you will need to identify all the requirements to satisfy the criterion. Finally, you need to create test cases to cover all the test requirements. Feel free to use more coverage criteria if you think that is necessary.
- 3) **Development of JUnit test cases:** Implement the test cases that you created in Task 2 above using the JUnit framework. This is similar to Question 1 of Assignment 3, except that we have a code base (instead of a single method) consisting of six classes to test.
- 4) **Running the test cases and analyzing test results:** Run all the JUnit test cases and observe test results. Is there any failed test case? Do you know why a test case failed? Also use JaCoCo (<https://www.jacoco.org/>), a Java code coverage tool, to generate a code coverage report, such as lines coverage, methods coverage, and branches coverage. The related coverage scores show how strong your test cases are.
- 5) **Test refinement:** Choose additional test coverage criteria and develop more test cases if necessary (e.g., to improve your JaCoCo scores).
- 6) **Project report** covering the following aspects:
 - a) Use at least three examples to explain how your group developed JUnit test cases to satisfy the coverage criterion/criteria that your team used. The explanation should be specific and similar to your answers to Question 1 of Assignment 3 (e.g., include graphs, TRs, test cases, and test paths). For each test case, explain what your input is and what your expected output is.
 - b) What problem(s) did you find in the code? For each problem, further explain how you found it (e.g., using which test case). What is your JaCoCo code coverage score? How did your group improve your coverage score? Include screenshots of JaCoCo test reports.
 - c) Describe your experience of designing JUnit test cases in this project and how your test coverage criteria helped (or hindered?) your test design. Putting everything together, what worked, what did not? What are your overall thoughts about software testing, test coverage criteria, or JUnit?

Your project will be graded based on the following aspects.

1. Code coverage score that JaCoCo reported.
2. The number and quality of JUnit test cases that you have developed.
3. The completeness and quality of your project report.
4. The problems that you have found in the Snake game code.
5. Project presentation as described below.

At the end, each group should submit two files to Cougar Course: (1) project report and (2) the zipped project file that includes your test cases for the Snake application.

Additionally, each group will give an in-class presentation about their project. The presentation counts 20% towards your project grade.